hcwave – Hospital AI Wave 1

Azure Infrastructure Prerequisites and Setup Guide

What must exist on Azure, and every step to put it in place, before the application is run.

Application	hcwave (WhatsApp patient assistant, reminders, report delivery, feedback, dashboard)
Deployment model	Bicep (subscription scope) → Entra apps → AKS add-ons → Helm → Logic Apps / Functions
Default regions	Central India (platform), South India (Azure OpenAI)
Environments	dev, staging, prod (one resource group each: `rg-hcw-<env>`)
Source of truth	`infra/bicep/main.bicep`, `scripts/*.sh`, `deploy/helm/hcwave`, `.github/workflows/deploy.yml`
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1. Overview

The application runs on **Azure Kubernetes Service (AKS)** and depends on a private-by-default landing zone: PostgreSQL, Blob Storage, Service Bus, Azure Communication Services (WhatsApp / SMS / Email), Azure OpenAI in Microsoft Foundry, AI Search, Content Safety, Translator, Key Vault, a container registry, and optional workflow engines (Logic Apps Standard, Durable Functions). A single Bicep deployment per environment creates almost everything. A small number of items **cannot** be automated and must be prepared by people *before* or *during* the rollout; these are marked **MANUAL** in this guide.

> **Order of work in one line:** accounts & permissions → provider registration & quotas → Entra group → Bicep infrastructure → Entra app registrations → WhatsApp channel → cluster add-ons → secrets → database bootstrap → application (Helm) → workflows → DNS/TLS → smoke test.

2. What gets created on Azure

All resources below are created by `infra/bicep/main.bicep` into resource group `rg-hcw-<env>` (the name prefix `hcw` is configurable, max 6 characters).

Area	Resource	Purpose
Networking	VNet `10.20.0.0/16` with subnets: AKS nodes, AGC, private endpoints, PostgreSQL (delegated), Logic Apps, Functions, APIM; NSGs	Private landing zone; internal job API on static IP `10.20.3.250`
Private DNS	Private DNS zones for Key Vault, ACR, PostgreSQL, Service Bus, Blob, Cognitive/OpenAI, Search, App Service sites	Name resolution for private endpoints
Identity	User-assigned managed identities: workload, ALB controller, Logic App, Functions	Passwordless access (workload identity, no keys in code)
Secrets	Key Vault (RBAC, private endpoint) + `app-secret-key`	CSI-mounted secrets; third-party tokens
Compute	AKS (workload identity, Key Vault CSI, Azure Policy, Defender, KEDA); system + user node pools	Runs API, worker, staff UI
Registry	Azure Container Registry	Images built with `az acr build`
Ingress	Application Gateway for Containers + WAF policy; Front Door Premium (prod/staging)	Public entry for API, reports, staff UI

Database	PostgreSQL Flexible Server (Entra auth, pgvector, private, delegated subnet)	Patients, appointments, audit, row-level security
Messaging	Service Bus namespace (Premium) with queues incl. `inbound-messages`, `delivery-status`, `report-events`	Async processing; KEDA scaling
Files	Storage account (reports; lifecycle delete after 1,825 days) + Event Grid to Service Bus	Report PDFs, signed-URL delivery
Channels	Azure Communication Services, Email Service + managed domain, Event Grid system topic	WhatsApp, SMS, Email; inbound events via Event Grid → Service Bus
AI	Microsoft Foundry / Azure OpenAI (`gpt-5-mini`, `gpt-5`, `text-embedding-3-large`), AI Search, Content Safety, Translator	Assistant, retrieval, safety, Malayalam/Hindi/English
Observability	Log Analytics, Application Insights, alert action group	Diagnostics for all resources
Workflows	Runtime storage, Logic App Standard, Function App (Durable) – VNet-integrated, private	Reminders / recurrence jobs / feedback orchestration
Resilience	Recovery Services / Backup vault and policies	Prod backup of PostgreSQL / AKS
Compliance	Defender for Cloud plans; Azure Policy (allowed locations: India, pod security, MCSB audit)	Data residency and security posture
Optional	APIM (AI gateway), Managed Grafana, Power BI Embedded, Purview, Data Factory, FHIR service, public DNS zone	Switched on by feature flags

3. Environment differences

Setting	dev	prod
AKS API server	Public	Private cluster
AKS user pool	1–3 nodes	3–8 nodes (Standard_D4ds_v5)
PostgreSQL	Burstable B2ms, no HA	GeneralPurpose D4ds_v5, zone HA, geo-redundant backup
Key Vault / ACR network	Public enabled	Public disabled (private endpoints only)
Front Door WAF	Off	On
APIM, Backup, Defender	Off	On

Purview, Data Factory	Off	On
AI Search SKU	basic	standard
Service Bus SKU	Premium in both (a cost to review for dev)	

> **Staging:** only `dev.bicepparam` and `prod.bicepparam` exist. Create `infra/bicep/parameters/staging.bicepparam` by copying `dev.bicepparam` and setting `environmentName = 'staging'` before running the staging steps.

4. Before you start – prerequisites

4.1 Azure subscription and permissions

- An active **Azure subscription** (one per environment is recommended; at minimum a dedicated one for prod) with a billing method that allows Premium SKUs.
- The deploying identity needs **Owner** on the subscription (or **Contributor + User Access Administrator**). The Bicep creates many RBAC role assignments and subscription-level Defender / Policy resources, so Contributor alone will fail.
- In Microsoft Entra ID: right to **create app registrations** and service principals (Application Developer or higher) and permission to **grant admin consent** (Cloud Application Administrator / Global Administrator) for the staff API and SPA.
- An Entra user able to create a **security group** (or an existing one).

4.2 Workstation or CI runner tooling

The scripts are Bash. On Windows use WSL2 (Ubuntu) or Git Bash, or run from a Linux CI runner.

Tool	Used by
Azure CLI (`az`) with Bicep (`az bicep install`)	all scripts, deployment
`jq`, `uuidgen`, `zip`, `git`, `make`	deploy-infra, entra-apps, deploy-workflows, deploy-app
`kubectl`, `helm` (v3), `kubelogin`	install-cluster-addons, deploy-app
`psql` (PostgreSQL client)	db-bootstrap

Sign in: `az login` then `az account set --subscription <id>`.

4.3 Network reachability (important for prod)

In prod, Key Vault, ACR, PostgreSQL, the AKS API server, Logic Apps and Functions are **private**. Steps that talk to them (`kv-secrets`, `db-bootstrap`, `deploy-app`, `deploy-workflows`) must run from a machine **inside or peered to the VNet** (self-hosted runner, jump box, VPN/ExpressRoute) or use `az aks command invoke`. Plan this host before prod rollout.

5. Azure-side preparation

5.1 Register resource providers (once per subscription)

```
for p in Microsoft.ContainerService Microsoft.ContainerRegistry Microsoft.KeyVault \
  Microsoft.DBforPostgreSQL Microsoft.ServiceBus Microsoft.EventGrid Microsoft.Storage \
  Microsoft.Communication Microsoft.CognitiveServices Microsoft.Search \
  Microsoft.Network Microsoft.ServiceNetworking Microsoft.Cdn Microsoft.ApiManagement \
  Microsoft.Web Microsoft.Logic Microsoft.OperationalInsights Microsoft.Insights \
  Microsoft.ManagedIdentity Microsoft.Authorization Microsoft.PolicyInsights \
  Microsoft.Security Microsoft.RecoveryServices Microsoft.DataProtection \
  Microsoft.KubernetesConfiguration Microsoft.Dashboard Microsoft.PowerBIDedicated \
  Microsoft.Purview Microsoft.DataFactory Microsoft.HealthcareApis; do
  az provider register --namespace $p; done
az provider list --query "[?registrationState!='Registered'].namespace" -o tsv   # should become
empty
```

5.2 Quotas and access approvals (request early – can take days)

Item	Requirement
Azure OpenAI / Foundry (South India)	Quota for `gpt-5-mini` (50), `gpt-5` (20) and `text-embedding-3-large` (50), Standard SKU, in the `aiLocation` region. Model availability varies by region; `gpt-5` may need an access request. Verify in the Foundry portal → Quotas. If a region lacks a model, change `HCWAVE_AI_LOCATION` or edit `openAiDeployments`.
Compute vCPUs (Central India)	Dsv5/Ddsv5 family vCPUs: prod needs up to 4 (system) + 8 × 4 (user) = 36+ (plus surge); dev ~16. Request an increase if short.
PostgreSQL Flexible Server	SKU availability and zone-redundant HA in Central India (check `az postgres flexible-server list-skus -l centralindia`).
Service Bus Premium, AI Search, APIM	Available in region; APIM and Premium Service Bus are costly – confirm budget.
Azure Communication Services	Advanced Messaging (WhatsApp) availability for your tenant; SMS number / sender (India requires DLT registration of sender ID and templates).

5.3 Entra ID group and principals (MANUAL)

1. Create a security group, e.g. `hcwave-admins`, add platform admins. It becomes the **AKS cluster admin** and the **PostgreSQL Entra administrator**.

   ```
   az ad group create --display-name hcwave-admins --mail-nickname hcwave-admins
   az ad group show --group hcwave-admins --query id -o tsv      # = HCWAVE_ADMIN_GROUP_ID
   ```

2. Get the object ID of the person/pipeline that will write Key Vault secrets (`HCWAVE_OPERATOR_PRINCIPAL_ID`, optional but needed for `kv-secrets.sh`).
3. Add the person running `db-bootstrap.sh` to the admin group.

5.4 CI/CD identity (only if using GitHub Actions)

1. Create an app registration / service principal for deployment and give it **Owner** (or Contributor + User Access Administrator) on the subscription.

2. Add a **federated credential** for your repo and each GitHub environment (`dev`, `staging`, `prod`) – OIDC, no stored secret.
3. Add the service principal to the `hcwave-admins` group.
4. In each GitHub environment set variables: `AZURE_CLIENT_ID`, `AZURE_TENANT_ID`, `AZURE_SUBSCRIPTION_ID`, `HCWAVE_ADMIN_GROUP_ID`, `HCWAVE_ADMIN_GROUP_NAME`, `HCWAVE_ALERT_EMAIL`, `HCWAVE_DOMAIN`, and after the Entra step: `STAFF_API_AUDIENCE`, `STAFF_API_SCOPE`, `ENTRA_SPA_CLIENT_ID`, `INTERNAL_API_AUDIENCE`, `FUNCTIONS_AUDIENCE`, `DEMO_HOSPITAL_CHANNEL_IDS`.

5.5 Decisions to make up front

Decision	Where it is used
Name prefix (≤ 6 chars, default `hcw`)	`HCWAVE_NAME_PREFIX`
Primary and AI region (defaults centralindia / southindia)	`location`, `HCWAVE_AI_LOCATION`
Ops alert e-mail	`HCWAVE_ALERT_EMAIL` (also APIM publisher in prod)
Public domain (e.g. `hospital.example`) – hosts `api.`, `reports.`, `staff.`	`HCWAVE_DOMAIN`; `HCWAVE_DNS_ZONE` if Azure hosts the zone
Feature switches (Front Door, APIM, Logic Apps, Functions, Backup, Defender, Policy, Grafana, Power BI Embedded, Purview, Data Factory, FHIR)	parameter files
Restrict AKS API by IP (`aksAuthorizedIpRanges`)	dev/staging with a public cluster

5.6 WhatsApp / Meta prerequisites (MANUAL, start early)

- A verified **Meta Business account** and a **WhatsApp Business Account** with a phone number not already registered on WhatsApp.
- Display name approval and business verification (can take days).
- Message **templates** prepared for approval (reminders, report links, feedback, handover). The list is in Appendix A of `docs/Hospital_AI_Wave1_As-Built_Solution_Architecture.docx` and `config/messaging/templates.yaml`.
- For e-mail: a custom sending domain (add DNS records for SPF / DKIM) if you do not use the Azure-managed domain. For SMS: sender ID / DLT registration, or an MSG91 account.

6. Step-by-step deployment

Run from the repository root (`hcwave/`) in a Bash shell. Replace `ENV` with `dev`, `staging` or `prod`. `make` targets wrap the scripts shown.

Step 1 – Sign in and select the subscription

```
az login
az account set --subscription "<subscription-id>"
az account show -o table
```

Step 2 – Export deployment variables

```
export HCWAVE_ADMIN_GROUP_ID="<entra-group-object-id>"
export HCWAVE_ADMIN_GROUP_NAME="hcwave-admins"
export HCWAVE_OPERATOR_PRINCIPAL_ID="<object-id-who-writes-secrets>"   # optional
export HCWAVE_ALERT_EMAIL="ops@yourhospital.example"
export HCWAVE_NAME_PREFIX="hcw"                    # optional
export HCWAVE_AI_LOCATION="southindia"             # optional
export HCWAVE_DNS_ZONE="hospital.example"          # optional (prod)
```

Step 3 – Preview the infrastructure (what-if)

```
make infra-whatif ENV=dev
# = scripts/deploy-infra.sh dev --what-if
```

Review that the resource group, AKS, PostgreSQL, OpenAI deployments and role assignments are listed with no errors (quota or region problems surface here).

Step 4 – Deploy the infrastructure

```
make infra ENV=dev
# = az deployment sub create --location centralindia --name hcwave-dev \
#     --template-file infra/bicep/main.bicep --parameters infra/bicep/parameters/dev.bicepparam
```

- First deployment takes **30–60 minutes**.
- Outputs are written to `deploy/generated/<env>.env` (AKS, ACR, Key Vault, PostgreSQL host, Service Bus namespace, storage URL, ACS endpoint, OpenAI / Search / Content Safety / Translator endpoints, identity client IDs, AGC IDs) and a Helm overlay `values-<env>.generated.yaml` is rendered. Do not commit these files.

> **Re-deployment caution:** `appSecretKey` defaults to `newGuid()`, so re-running the Bicep deployment can rotate the HMAC key used for cookies and OTP hashing. Pass an explicit stable value for production re-deployments.

Step 5 – Create the Entra app registrations

```
make entra ENV=dev UI_URL=https://staff.<domain>
# = scripts/entra-apps.sh dev https://staff.<domain>
```

This creates `hcwave-<env>-staff-api` (app roles FrontOffice, PatientRelations, UnitHead, Admin; scope `access`), `-internal-api`, `-functions` and the SPA `-staff-ui`. Then:

1. Append the printed values (`STAFF_API_AUDIENCE`, `STAFF_API_SCOPE`, `ENTRA_SPA_CLIENT_ID`, `INTERNAL_API_AUDIENCE`, `FUNCTIONS_AUDIENCE`) to `deploy/generated/<env>.env`.
2. **MANUAL:** grant admin consent to the SPA for the staff API scope.
3. **MANUAL:** in Enterprise applications → `hcwave-<env>-staff-api` assign users/groups to the app roles.
4. **MANUAL:** add a `hospital_ids` claim (directory extension or claims-mapping policy) listing each user's hospital IDs – used for row-level security.
5. Make sure the audiences match the Bicep parameters `internalApiAudience` / `functionsAuthAudience` (defaults are `api://hcwave-internal-api` and `api://hcwave-functions`; the script creates `api://hcwave-<env>-...`). Update the parameters and re-run Step 4 if they differ.
6. Re-render values: `scripts/render-values.sh <env>` (also done by Step 9).

Step 6 – Connect WhatsApp in Azure Communication Services (MANUAL)

1. Azure portal → the ACS resource `hcw-<env>-acs` → **Advanced Messaging** → **WhatsApp** → connect the WhatsApp Business Account through the Meta embedded sign-up.
2. Note the **channel registration ID**; set it as `DEMO_HOSPITAL_CHANNEL_IDS` (and in the hospital config under `config/hospitals/`).
3. Submit the message templates in the portal / Meta Business Manager and wait for approval.
4. Confirm the Event Grid system topic `hcw-<env>-acs-events` delivers to Service Bus queues `inbound-messages` and `delivery-status`.
5. E-mail: verify the sender domain (managed domain works out of the box). SMS: purchase/assign a number or configure MSG91; set `ACS_SMS_FROM`.

Step 7 – Install cluster add-ons

```
make addons ENV=dev
# = scripts/install-cluster-addons.sh dev [--with-cert-manager] [--with-backup]
```

Installs the **ALB controller** (Application Gateway for Containers) with the ALB managed-identity client ID, optionally **cert-manager** (Let's Encrypt for the gateway) and the **AKS Backup extension** (prod), and creates the `hcwave` namespace. Needs `kubelogin`; on a private cluster run from inside the VNet.

Step 8 – Store third-party secrets in Key Vault (only those you use)

```
META_ACCESS_TOKEN=... META_APP_SECRET=... META_VERIFY_TOKEN=... MSG91_AUTH_KEY=... \
HIS_API_TOKEN=... scripts/kv-secrets.sh dev
```

Most Azure services use managed identity and need no secret. Optional secrets: Meta Cloud API tokens (only if bypassing ACS), MSG91, Claude via Foundry (`ANTHROPIC_FOUNDRY_API_KEY`), hospital-system (HIS) API token, `AZURE_OPENAI_API_KEY`. Add each name to `secrets.azureKeyVault.objects` in the Helm values. The operator needs the Key Vault Secrets Officer role (granted via `HCWAVE_OPERATOR_PRINCIPAL_ID`) and network access to the vault.

Step 9 – Bootstrap the PostgreSQL role (once per environment)

```
make db-bootstrap ENV=dev
# = scripts/db-bootstrap.sh dev
```

Run as a member of the admin group from a host with VNet access (or `az aks command invoke`). It creates the Entra-authenticated role for the workload identity (`pgaadauth_create_principal`), grants schema rights, makes it database owner (so Alembic migrations run as it) and enables the `vector` extension. No database password is used.

Step 10 – Build images and deploy the application

```
make deploy ENV=dev
# = scripts/deploy-app.sh dev [tag]
#   az acr build hcwave/backend and hcwave/staff-ui, render values, helm upgrade --install --wait
```

Verify: `kubectl -n hcwave get pods` – API, worker and staff UI pods should be Running; migrations apply on start.

Step 11 – Deploy Logic Apps and Durable Functions

```
make workflows ENV=dev
# = scripts/deploy-workflows.sh dev
```

Zip-deploys `workflows/logic-apps` and `services/functions`. Both are private; run from the VNet or per your policy.

Step 12 – DNS and TLS

1. Create DNS records for `api.<domain>`, `reports.<domain>`, `staff.<domain>` pointing (CNAME) to the Front Door host (`FRONT_DOOR_HOST`) when Front Door is on, otherwise to the AGC frontend FQDN. If you set `dnsZoneName`, the Bicep creates the Azure DNS zone and records – delegate the domain's name servers to it at your registrar.
2. Issue certificates (cert-manager from Step 7, or Front Door managed certificates).
3. Set the WhatsApp / event webhook URLs if any provider calls the public API (`PUBLIC_BASE_URL` = `https://api.<domain>`).

Step 13 – Smoke test

```
scripts/smoke-test.sh https://api.<domain>
```

Checks `/healthz` (liveness) and `/readyz` (database reachable). Then send a WhatsApp message to the connected number and confirm a reply; sign in to the staff UI with a role-assigned user.

7. Verification checklist

☐ Resource group `rg-hcw-<env>` exists; deployment `hcwave-<env>` is Succeeded.
☐ All 3 OpenAI deployments Succeeded in the Foundry resource.
☐ AKS nodes Ready; ALB controller pods Running in `azure-alb-system`.
☐ Private endpoints Approved and private DNS zones linked to the VNet.
☐ PostgreSQL: workload identity role exists; `vector` extension enabled; `/readyz` returns OK.
☐ Service Bus queues receive ACS events (inbound-messages, delivery-status).

☐ Key Vault contains `app-secret-key` and any optional secrets; pods mount them via CSI.
☐ Entra: app roles assigned, admin consent granted, `hospital_ids` claim issued.
☐ WhatsApp templates approved; test message round-trip works.
☐ Prod only: Front Door WAF active, Defender plans on, Backup policies protecting PostgreSQL, alert e-mails received.

8. Common issues

Symptom	Likely cause / fix
Deployment fails on role assignment	Deployer lacks User Access Administrator / Owner.
OpenAI deployment fails (quota / model not found)	No quota or model not offered in `aiLocation`; request quota or change region / model list.
Provider not registered	Run Section 5.1 and wait until all show Registered.
Policy blocks a resource	Allowed locations are India + global only; keep resources in those regions.
`db-bootstrap` / `kv-secrets` cannot connect	Private endpoints; run from a VNet host or use `az aks command invoke`; ensure you are in the admin group.
Pods cannot pull images	ACR public access disabled in prod and runner is outside VNet, or kubelet lacks AcrPull (granted by Bicep).
Staff UI sign-in fails	Redirect URI does not match `UI_URL`, admin consent missing, or user has no app role.
Internal API calls from Logic Apps / Functions rejected	Audience mismatch (Step 5, item 5) or `INTERNAL_ALLOWED_CLIENT_IDS` missing identity client IDs.

9. Estimated time and effort

Phase	Typical duration
Subscription, permissions, providers	0.5 day
Quota and model approvals	1–5 days (elapsed, parallel)
Meta business verification and WhatsApp templates	3–10 days (elapsed, parallel)
Bicep infrastructure deployment	30–60 minutes
Entra, add-ons, secrets, DB bootstrap, app, workflows	0.5–1 day
DNS, TLS, smoke test, go-live checks	0.5 day

Prepared from the repository source (Bicep modules, deployment scripts, Helm values and CI workflow). Resource names, SKUs and defaults reflect the files at the time of writing; the Bicep parameter files remain authoritative.